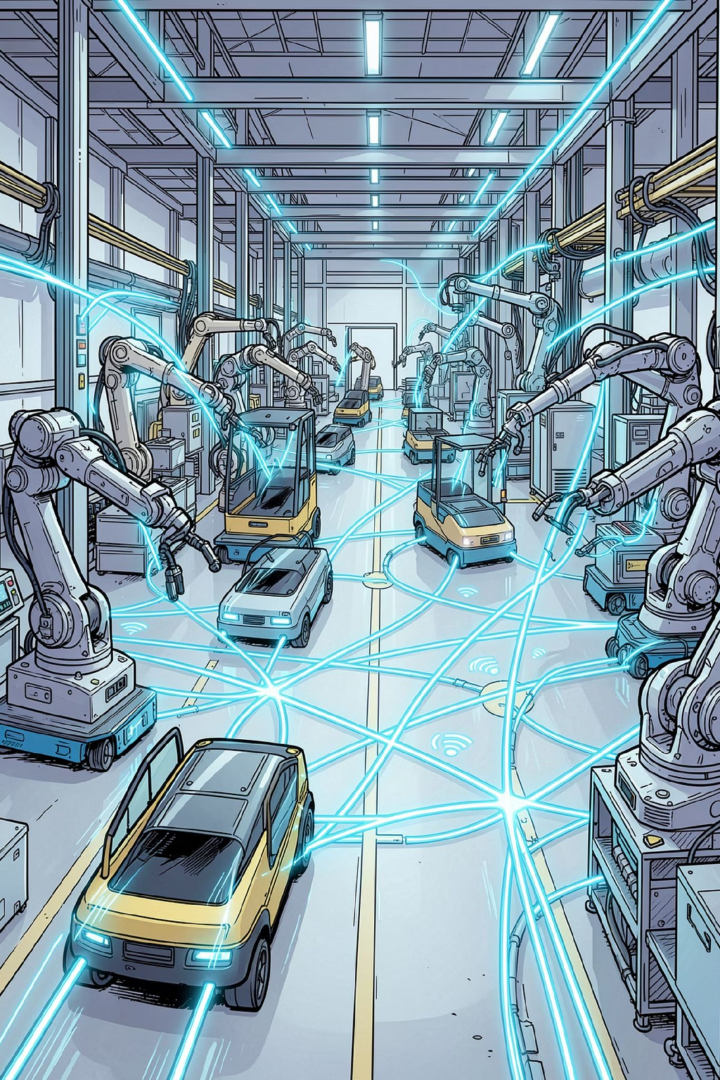
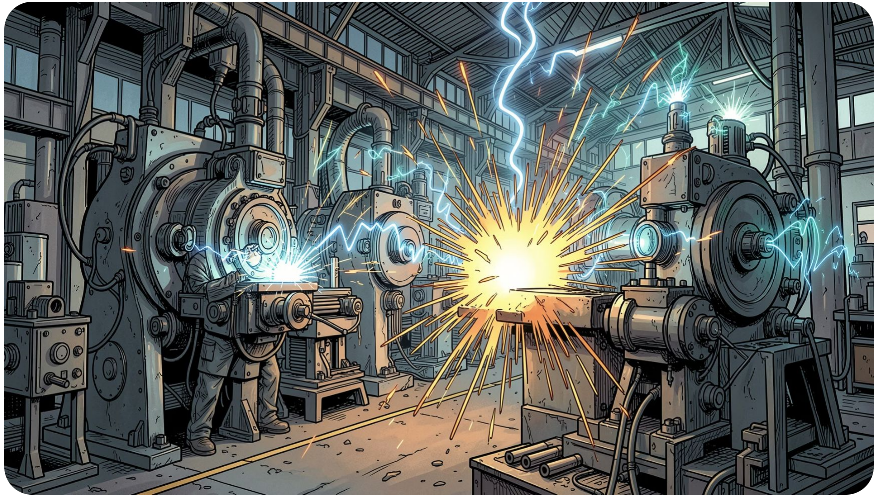


INDUSTRY 4.0

Why Wireless Matters in Industry 4.0

Industry 4.0 demands flexible, mobile, and scalable communication. Wired systems like PROFINET offer stable timing but are expensive to install and hard to reconfigure. Wireless enables AGVs, mobile sensors, and remote monitoring — but the real question is: **which wireless standard fits which industrial task?**

The Industrial Wireless Environment



Why Factories Are Hard on Wireless

Metal structures, motors, and moving machines create EMI, multipath fading, and unstable signal quality. Shared 2.4 GHz spectrum adds congestion; cybersecurity is critical.

Three Standards Under Evaluation

Wi-Fi / IEEE 802.11

High-bandwidth mobile communication

Zigbee / IEEE 802.15.4

Low-power sensor networks

5G New Radio

Real-time control & private networks

Evaluated on latency, throughput, reliability, energy use, and deployment complexity.

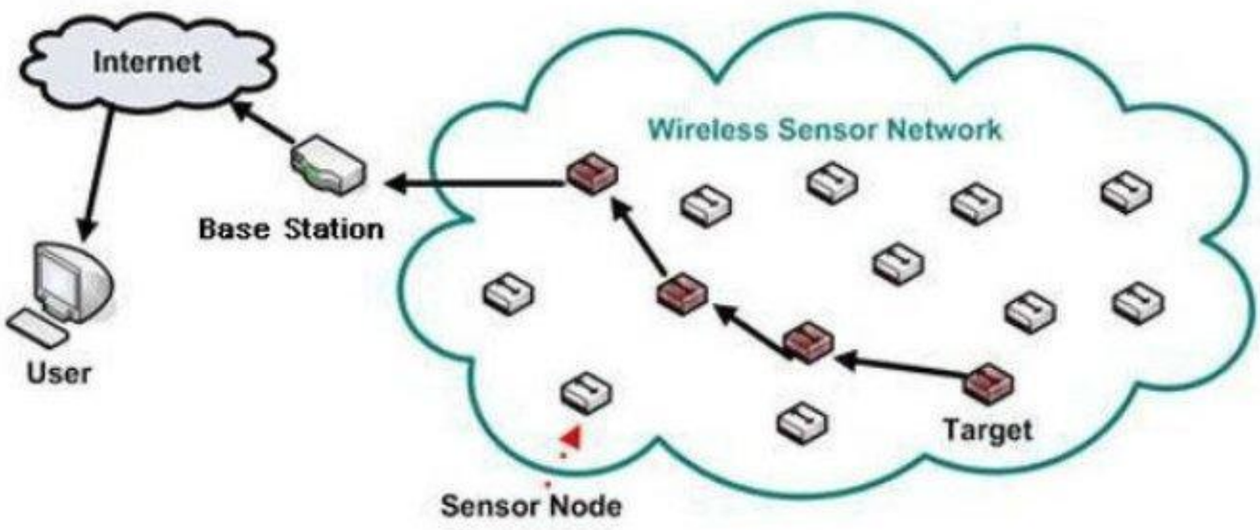
Literature Review

Industrial Wireless Communication for Industry 4.0

2.1 Early Wireless Sensor Networks (WSN)

Foundation	WSN established the foundation of industrial wireless communication, based on IEEE 802.15.4 standards.
Main Goal	Collect and transmit data wirelessly with low power consumption, allowing devices to run on batteries for years.
Key Challenges	Unstable delays, electromagnetic interference from machines, and signal loss in metallic factory environments.
Significance	Despite the challenges, WSN shaped the design principles still used in modern industrial wireless systems today.

2.1 WSN Architecture Overview



Typical WSN architecture: sensor nodes relay data through the network to a base station and the internet.

2.2 Industrial Wi-Fi Evolution

802.11b/g	High variable latency - not suitable for closed-loop industrial control.
802.11e	QoS (EDCA) introduced traffic prioritisation, improving reliability.
802.11n	MIMO antennas enabled spatial multiplexing and increased data rates.
Wi-Fi 6	OFDMA and TWT reduced latency to ~5 ms - suitable for high-bandwidth industrial applications.

2.2 Industrial Wireless Connectivity

Wireless connectivity links across an industrial network routers, sensors, and IoT-enabled devices.

2.3 5G and the URLLC Paradigm

< 1 ms Latency for motion control	**99.9999%** Reliability (URLLC)	**70%+** New private wireless spend on standalone 5G (2025)

- 3GPP Release 15 (2018): 5G NR air interface standardised.
- Release 16 (2020): mini-slot scheduling and enhanced HARQ added for URLLC.
- Release 16 also defines 5G as a TSN bridge for end-to-end deterministic delivery.
- Many industries are now adopting private 5G networks for real-time control tasks.

2.4 TSN Integration - Towards Wireless Determinism

What TSN Provides

- Credit-Based Shaping (802.1Qav)
- Time-Aware Shaping (802.1Qbv)
- Cyclic Queuing and Forwarding (802.1Qch)
- Sub-microsecond time synchronisation

Key Impact

- Wireless becomes as reliable as wired PROFINET / EtherCAT.
- Wired and wireless robot axes driven by the same scheduler.
- IEEE actively extending TSN standards to Wi-Fi (802.11).

Emerging Trends

Wi-Fi 7 · Private 5G · Wireless TSN · 6G Outlook

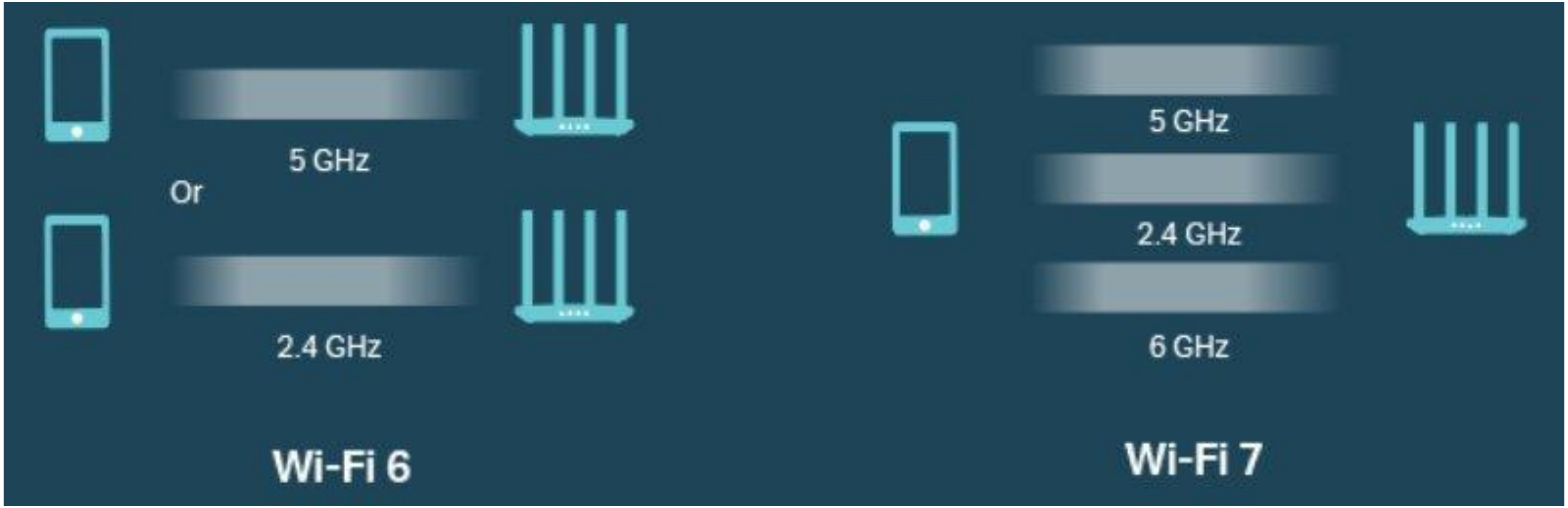
5.1 Wi-Fi 7 — Multi-Link Operation (MLO)

2.4 GHz	5 GHz	6 GHz

All three frequency bands used simultaneously

- MLO provides packet-level diversity — a dropped packet on one band is recovered on another without retransmission delay.
- IEEE 802.11be (Wi-Fi 7) is finalised. Draft 1.0 of Wi-Fi 8 (802.11bn) completed August 2025.
- Wi-Fi 8 targets deterministic performance and reliability rather than just higher throughput.

5.1 Wi-Fi 6 vs Wi-Fi 7 – Frequency Bands



Wi-Fi 6 uses 2.4/5 GHz bands, while Wi-Fi 7 adds the 6 GHz band for simultaneous multi-link operation.

5.2 Private 5G Campus Networks

- Companies build and manage their own 5G network instead of using public infrastructure.
- Full control over security, service levels, and spectrum allocation.
- Optimised for deterministic, high-density industrial environments such as factories.
- Complements industrial Ethernet and Wi-Fi in new (greenfield) factory deployments.

5.3 Wireless Time-Sensitive Networking

- TSN ensures data arrives at the exact required time - critical for real-time control.
- Closes the "last mile" gap: wireless now matches wired PROFINET and EtherCAT guarantees.
- A single TSN scheduler can drive both wired and wireless robot axes at the same time.

Wireless Standards: Architecture and Protocol Stack

Wi-Fi (802.11) · Zigbee (802.15.4) · 5G New Radio

3.1 IEEE 802.11 (Wi-Fi)

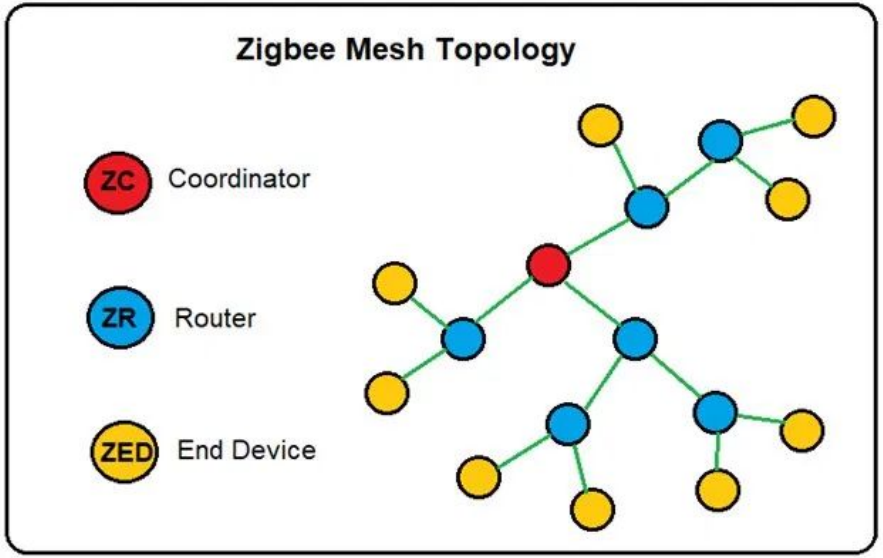
- Operates on 2.4, 5 and 6 GHz bands
- CSMA/CA for channel access
- Wi-Fi 6 adds OFDMA and TWT
- 802.11r enables fast roaming for AGVs
- Throughput up to 9.6 Gbps

PROS High speed, huge ecosystem, native TCP/IP

CONS Unreliable under load, shared spectrum

3.2 IEEE 802.15.4 / Zigbee

- Low-power wireless WPAN standard
- Runs at 2.4 GHz, up to 250 kbps
- Three roles: Coordinator, Router, End Device
- Supports star, tree and mesh topologies
- Battery life of 2 to 5 years

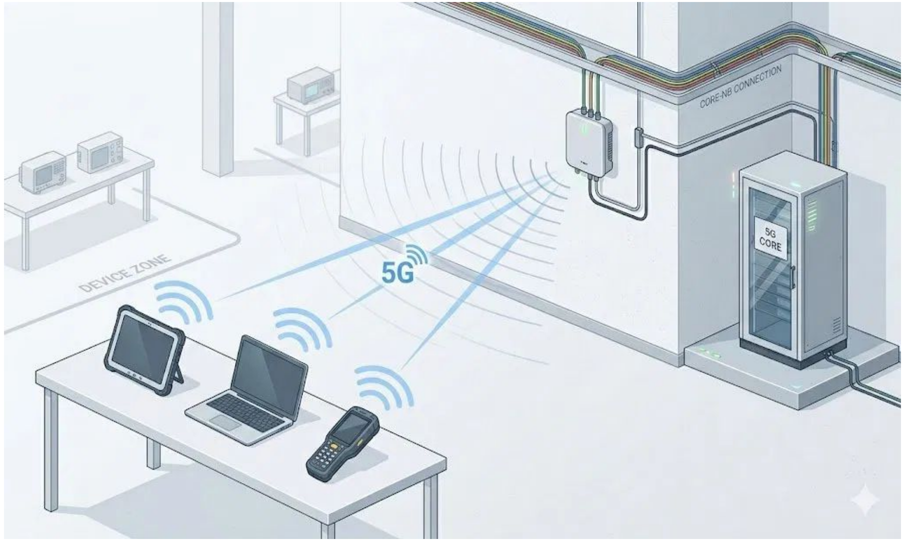


PROS Long battery life, low cost, self-healing mesh

CONS Low data rate, short range of 10 to 100 m

3.3 5G New Radio (NR)

- Standardised by 3GPP, uses licensed bands
- Three classes: eMBB, mMTC, URLLC
- URLLC reaches under 1 ms at 99.999%
- Private campus networks kept on site
- Acts as a TSN bridge from Release 16



PROS Licensed spectrum, deterministic, private networks

CONS Costly, complex, power-hungry

3.4 Standards at a Glance

	Wi-Fi	Zigbee	5G NR
Data rate	9.6 Gbps	250 kbps	Multi-Gbps
Latency	~5 ms	15 to 30 ms	< 1 ms
Range	~100 m	10 to 100 m	Wide / campus
Power use	High	Very low	High
Best for	High bandwidth	Sensors	Real-time control

Takeaway: no single standard fits every need, so factories mix all three by use case.

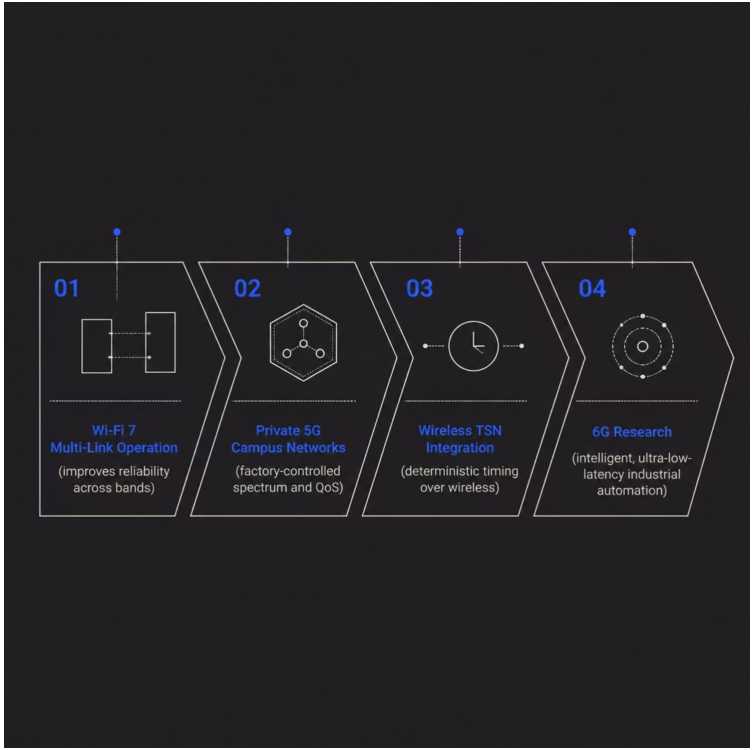
No Single Standard Fits Everything

The core finding: **each standard excels in a different domain.** Smart factories require a hybrid wireless architecture.

Standard	Best Suited For	Main Limitation
Wi-Fi 6/7	High-bandwidth mobile tasks: AGVs, machine vision, video monitoring	Shared spectrum, variable reliability
Zigbee	Low-power sensors: environmental monitoring, asset tracking	Low data rate
5G NR	Real-time control via URLLC in private campus networks	Higher cost & complexity

Key Takeaway: The optimal solution combines all three — matching each standard to its ideal use case.

Future Direction: Toward Reliable Wireless Factories



Challenges Still to Solve

- EMI and metal reflections
- 2.4 GHz band congestion
- Cybersecurity risks
- Deterministic timing gaps

Wi-Fi 7, private 5G, and Wireless TSN are closing the gap with wired systems. **Wireless won't replace every cable — but it will become central to flexible, mobile Industry 4.0 factories.**